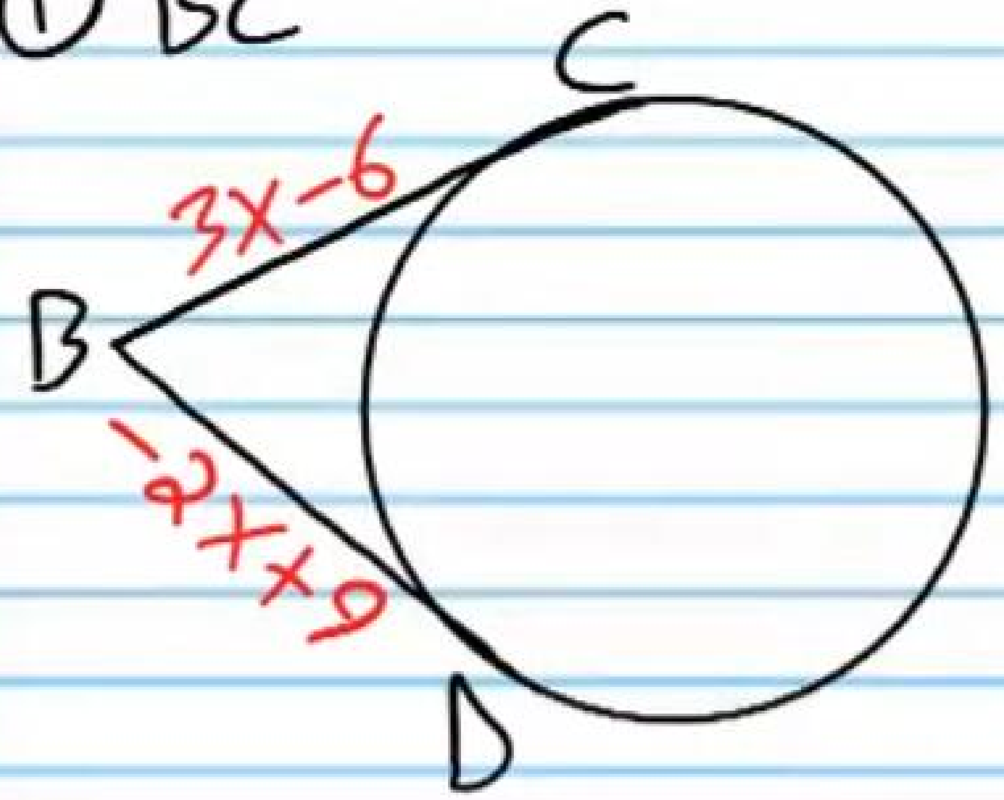
Name
Date
Course

Practice Geometry Test V (Version 2)

I) Write your name, the date, and the course title (i.e. Geometry).
II) Show work when necessary.
III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
IV) You will need a calculator for some problems on this test. When pi is required, use the π button on your calculator.
V) Graphing paper may be useful for this test.
VI) Each problem is worth four points for a total of 100 points.

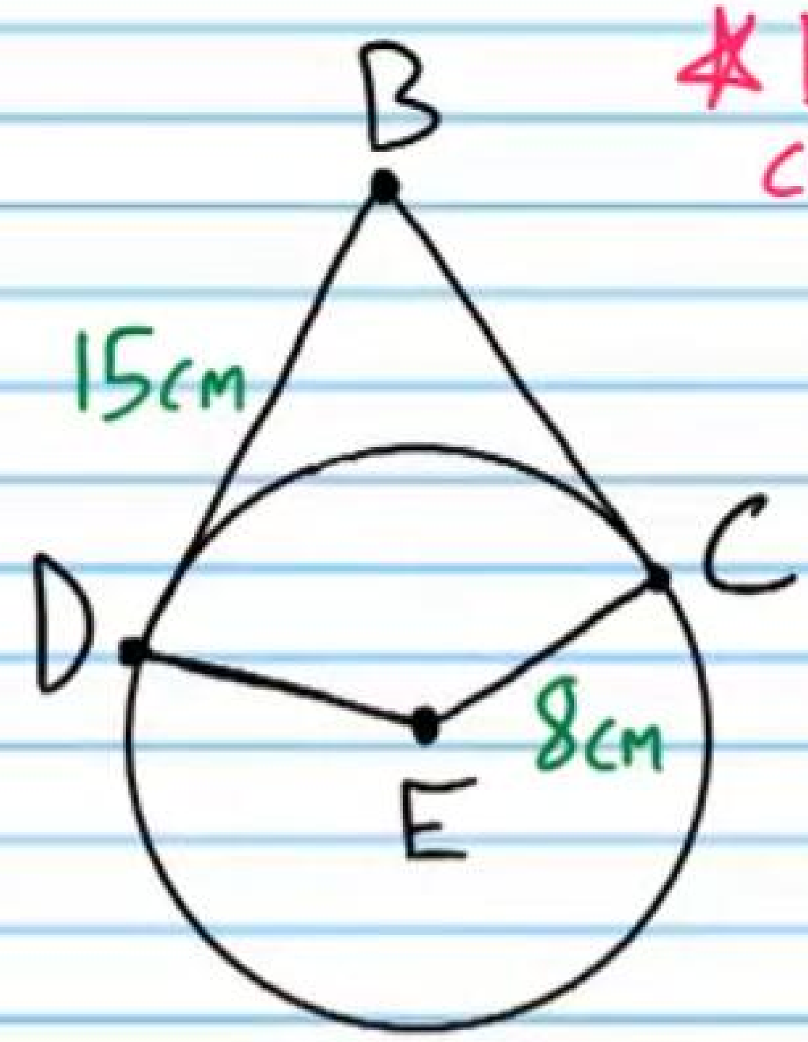
If C and D are points of tangency, find the following values.

① BC



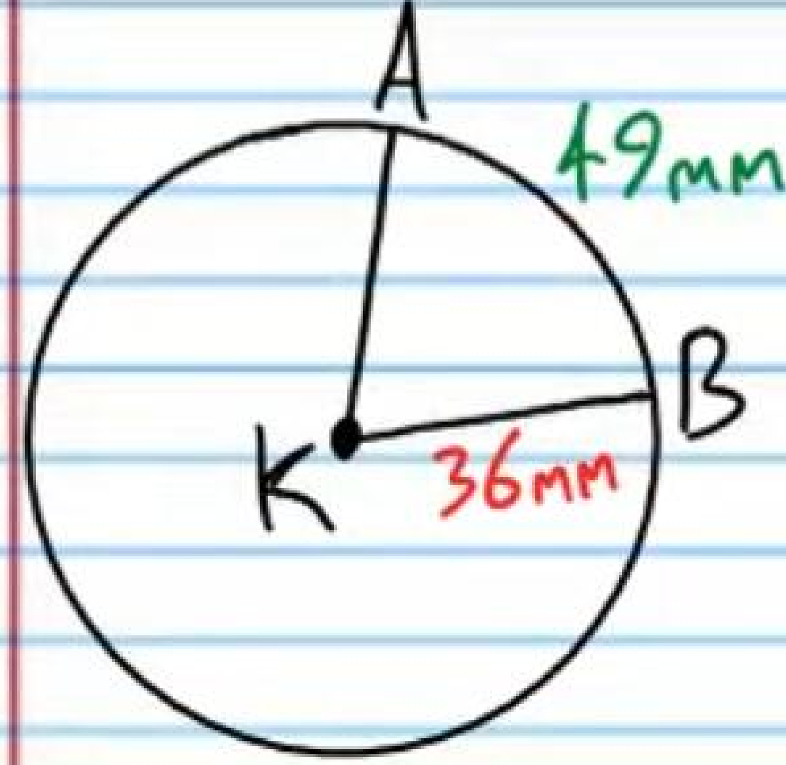
② i) m∠D & ii) EB

*E is the center.

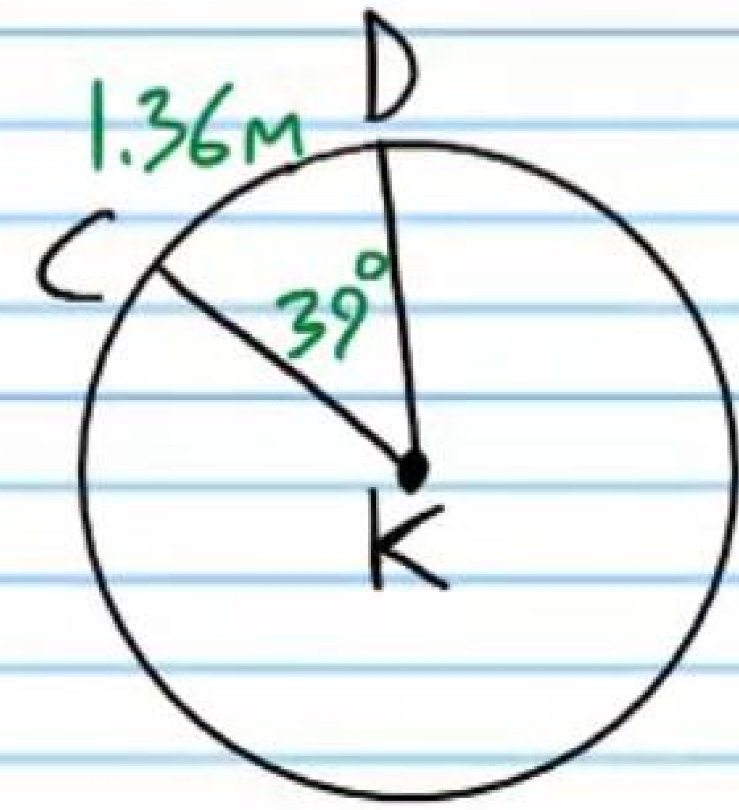


If K is the centers of the following circles, find the values below. If a problem requires the pi value, use the π button on your calculator. Round your answers to the nearest tenth if necessary.

③ m∠AKB

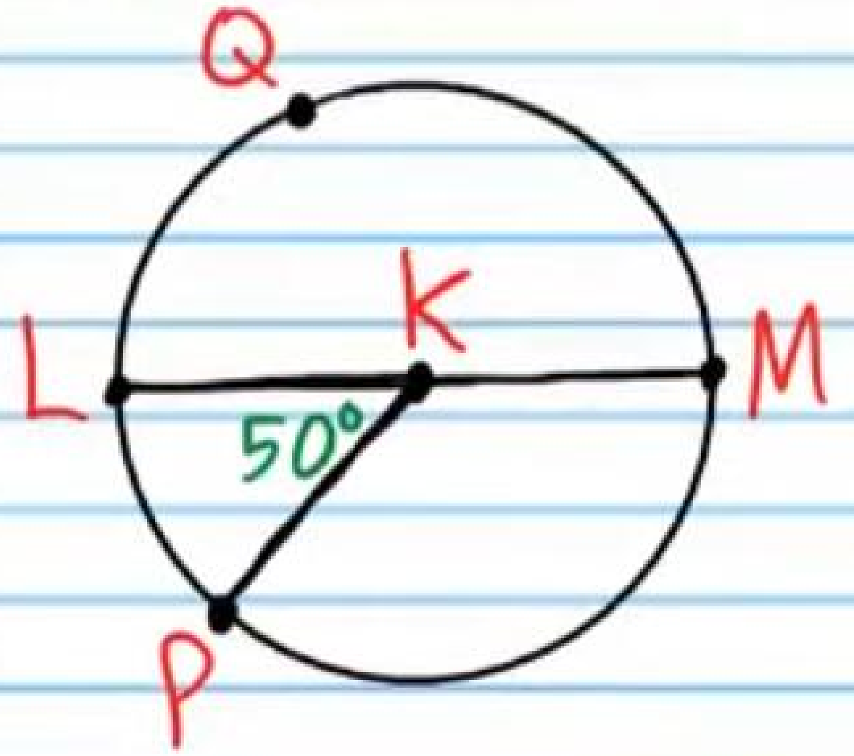


④ Circumference

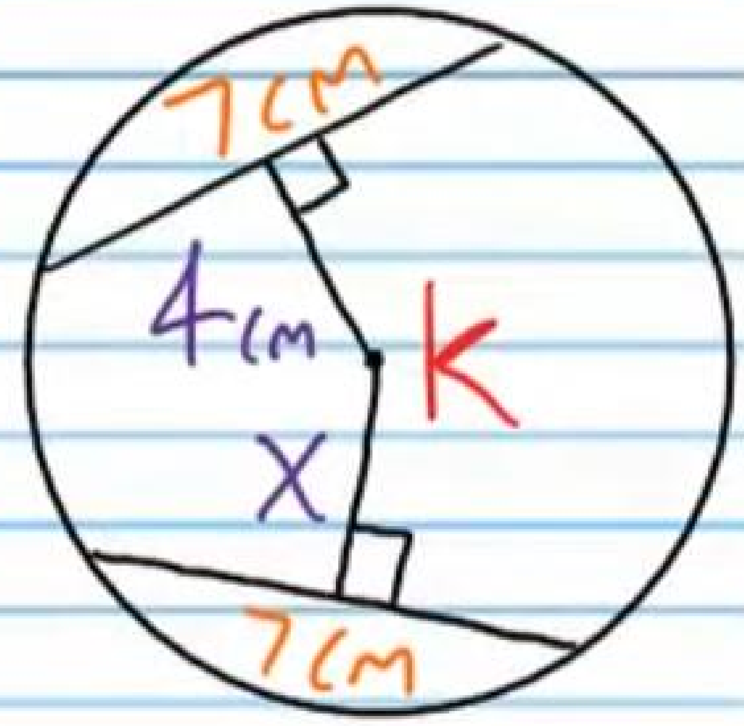


5. i) m$\widehat{LQM}$ ii) m$\widehat{LP}$

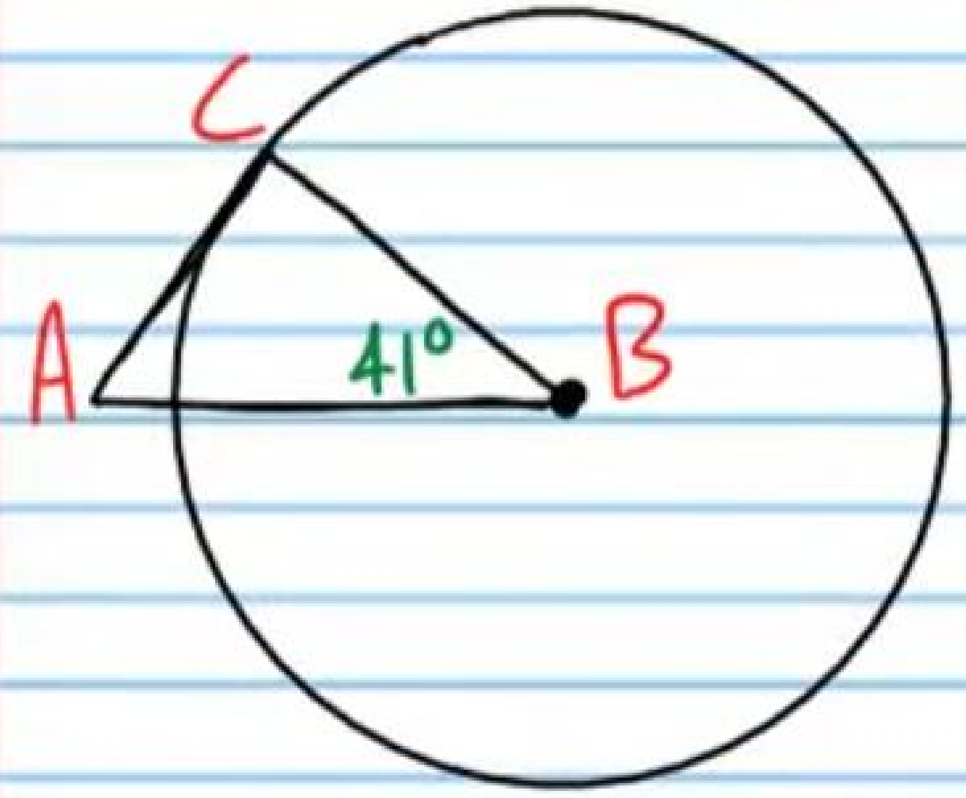
★ $\overline{LM}$ is a diameter.



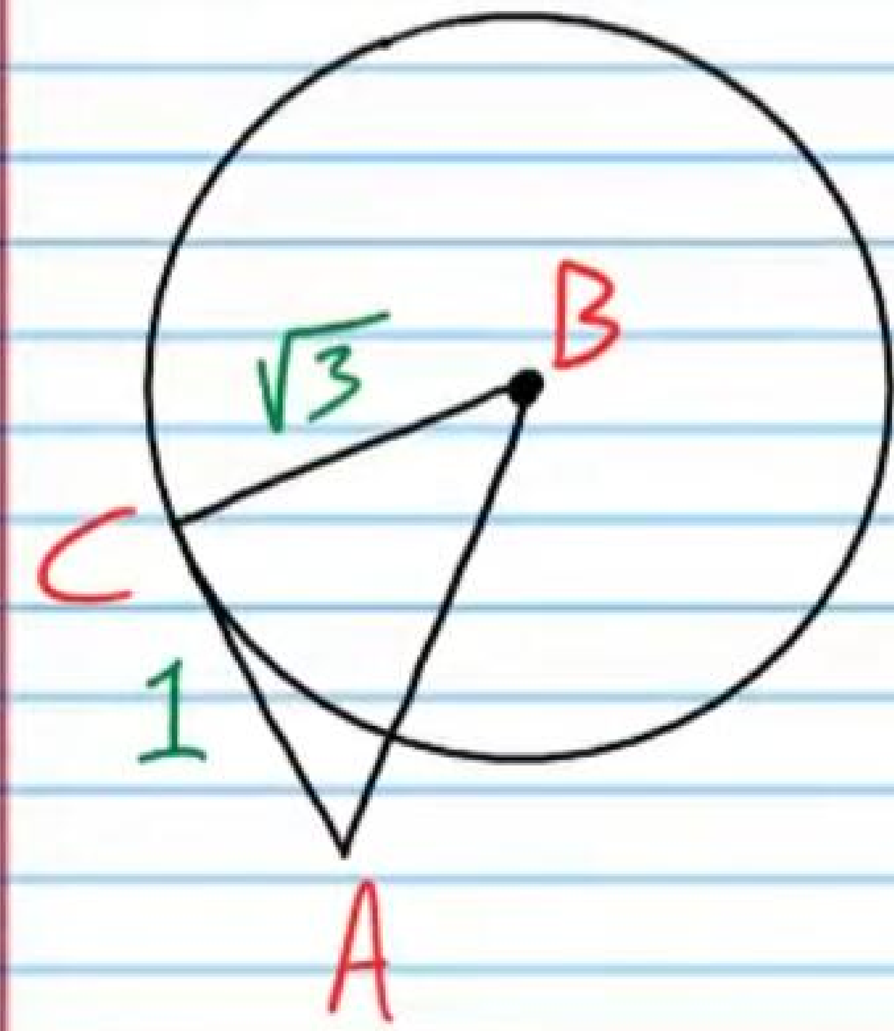
6. X



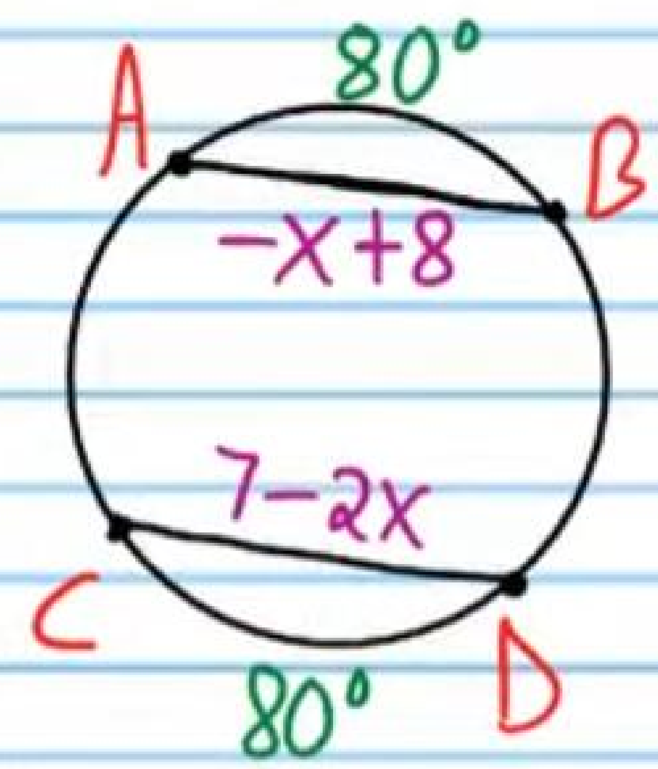
7. If the line containing $\overline{AC}$ is tangent to ⊙B at C, find i) m∠C and ii) m∠A.



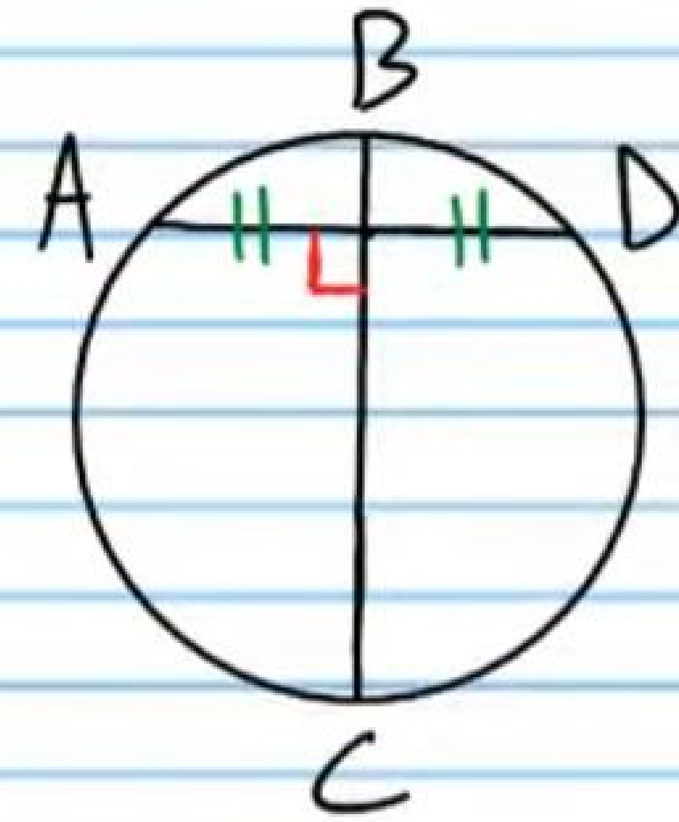
8. If $\overline{BC}$ is a radius, and AB = 2.03, is the line containing $\overline{AC}$ tangent to ⊙B? Show the calculations you use to determine tangency.



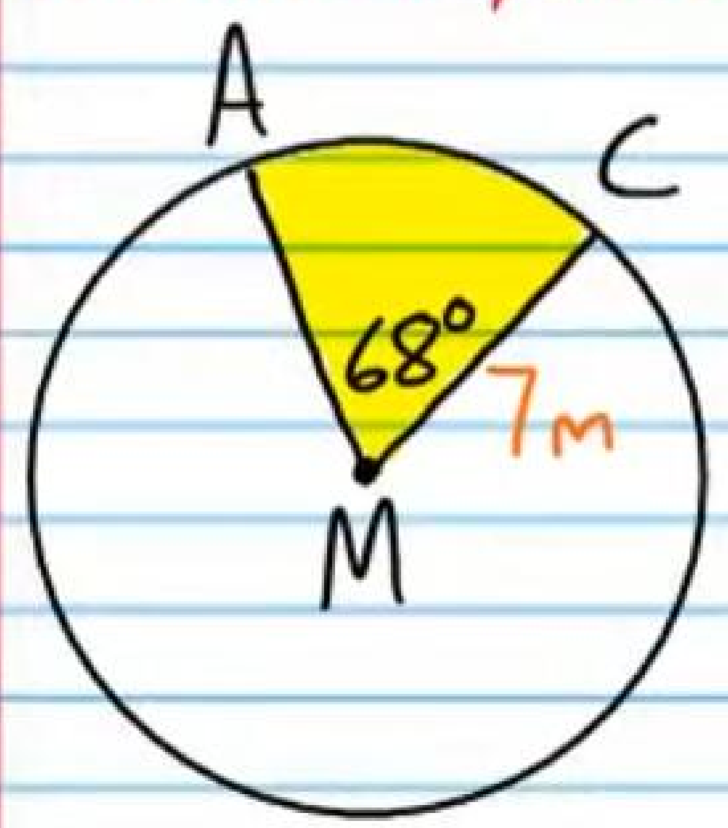
9. Find X.



10. Is $\overline{BC}$ a diameter? Explain.

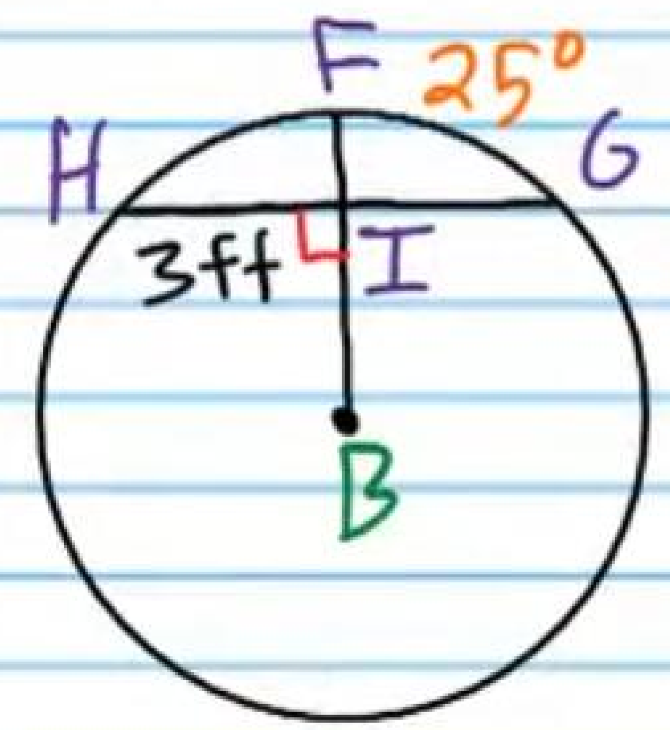


(11) If M is the center of the circle below, find the area of sector AMC. You must use the π button on your calculator. Round to the nearest hundredth.

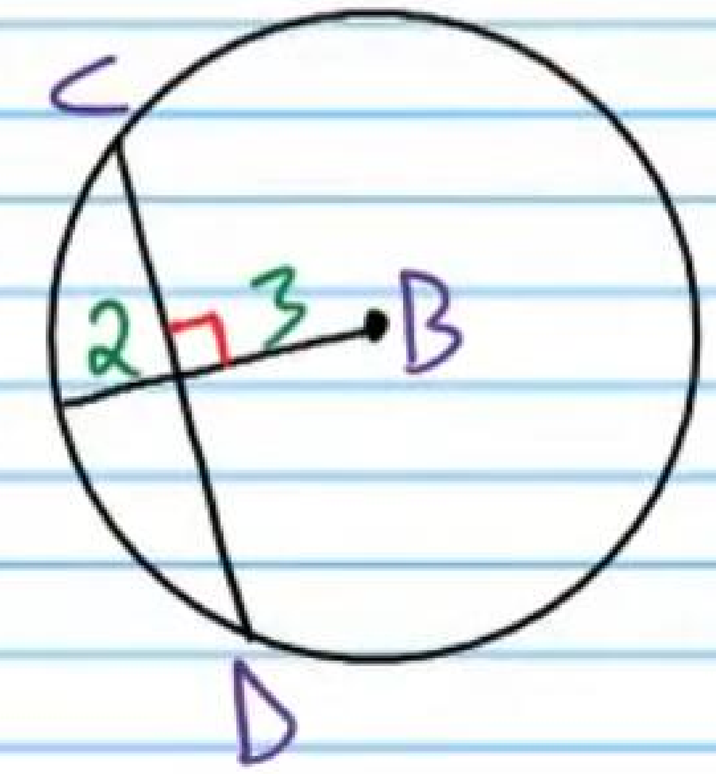


If B is the center of the circles below, find the following values.

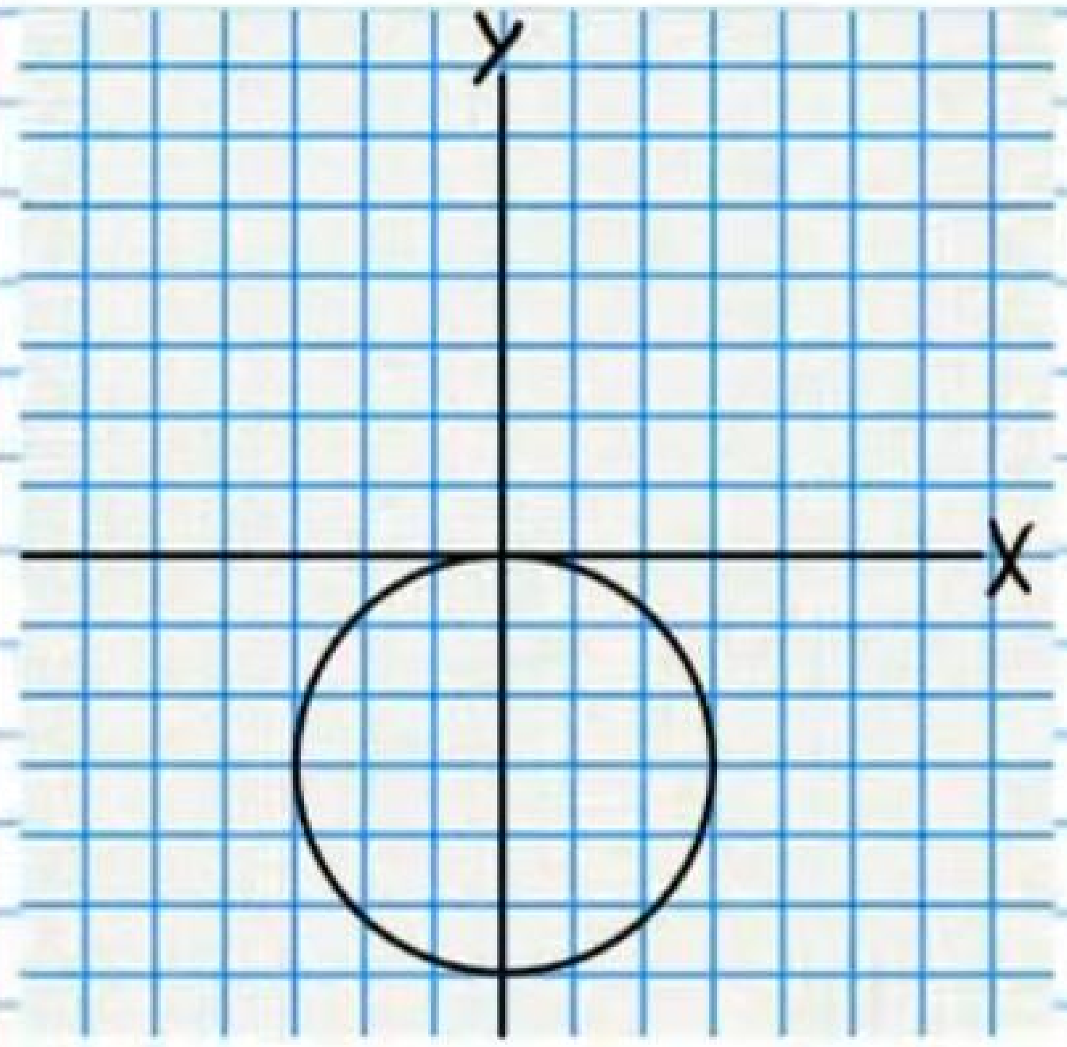
(12) i) $m\widehat{HG}$ & ii) HG



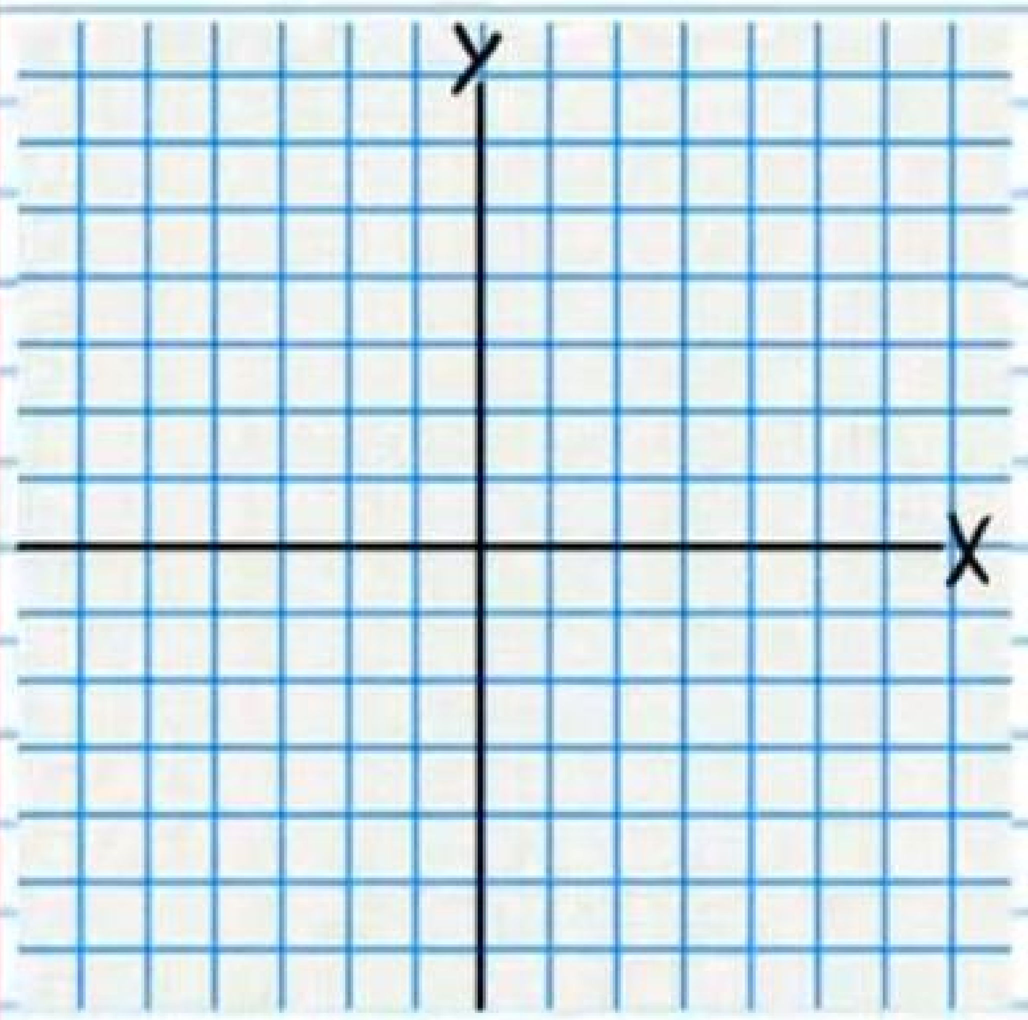
(13) CD



(14) Write an equation of the following circle.

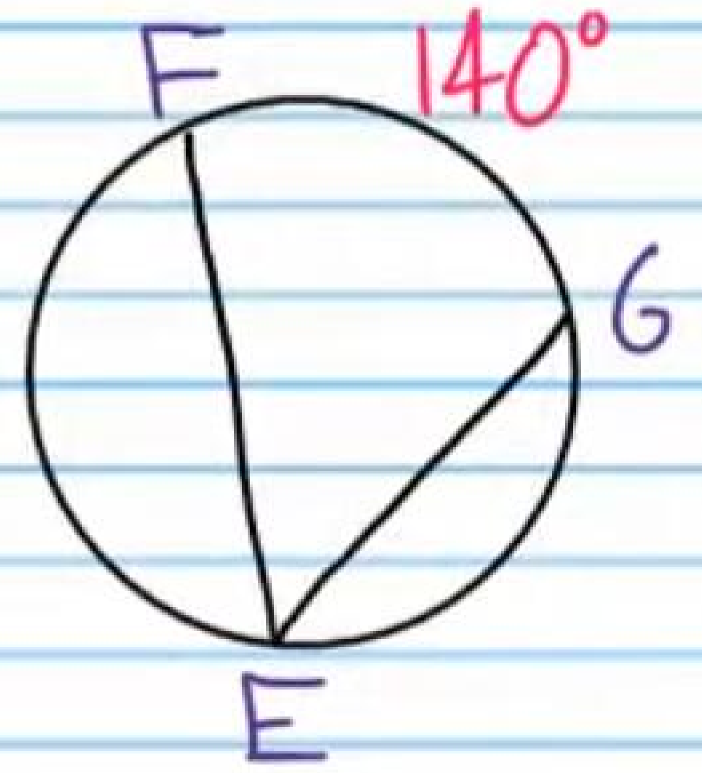


(15) Graph the following equation: $(x+3)^2+(y+2)^2=1$

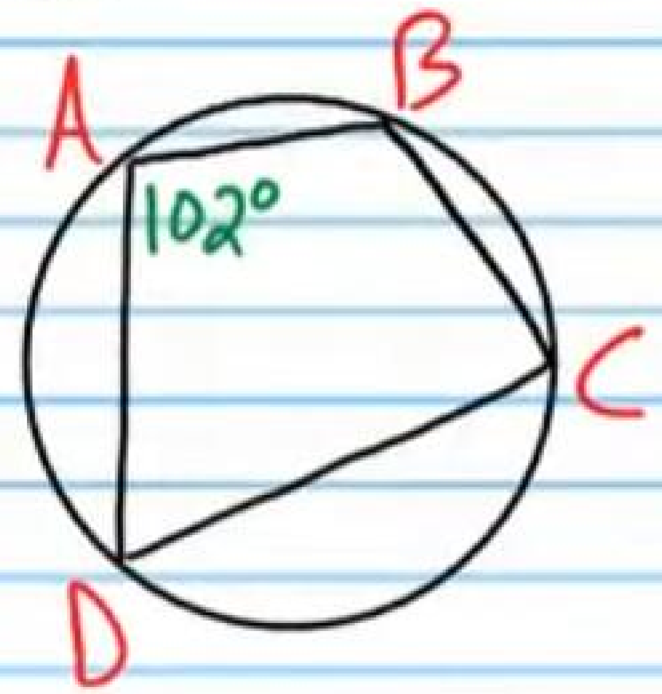


(16) Find an equation of the circle containing a diameter with endpoints (7,0) and (5,2).

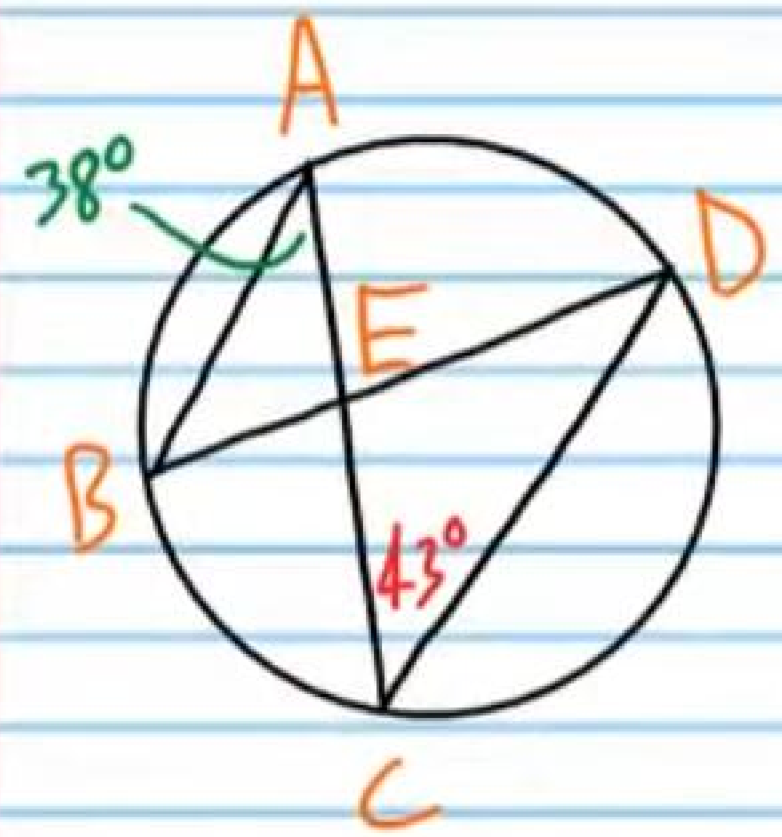
(17) Find m∠E.



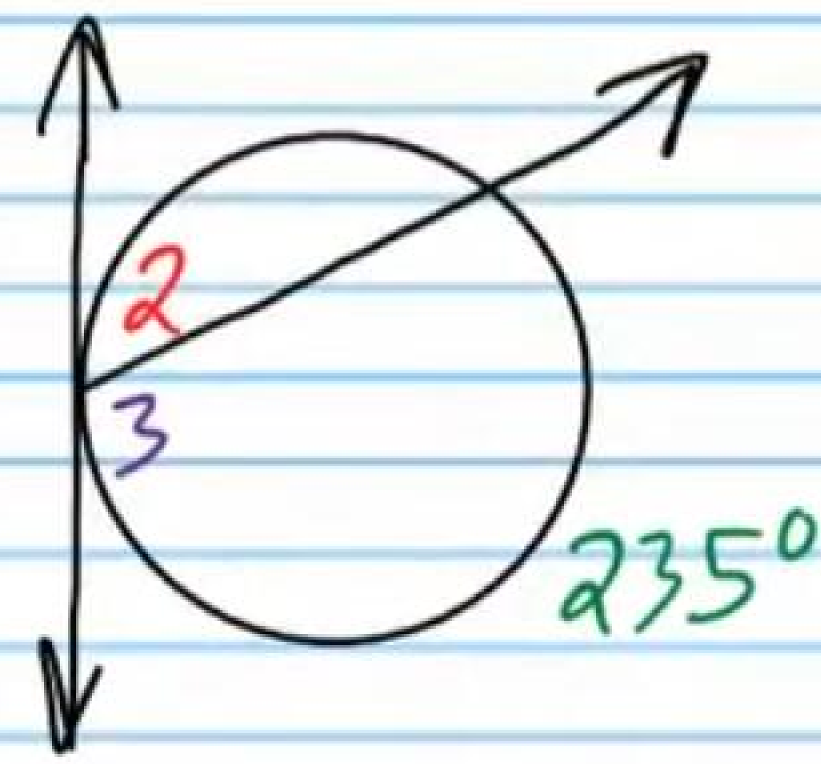
(18) Find m∠C.



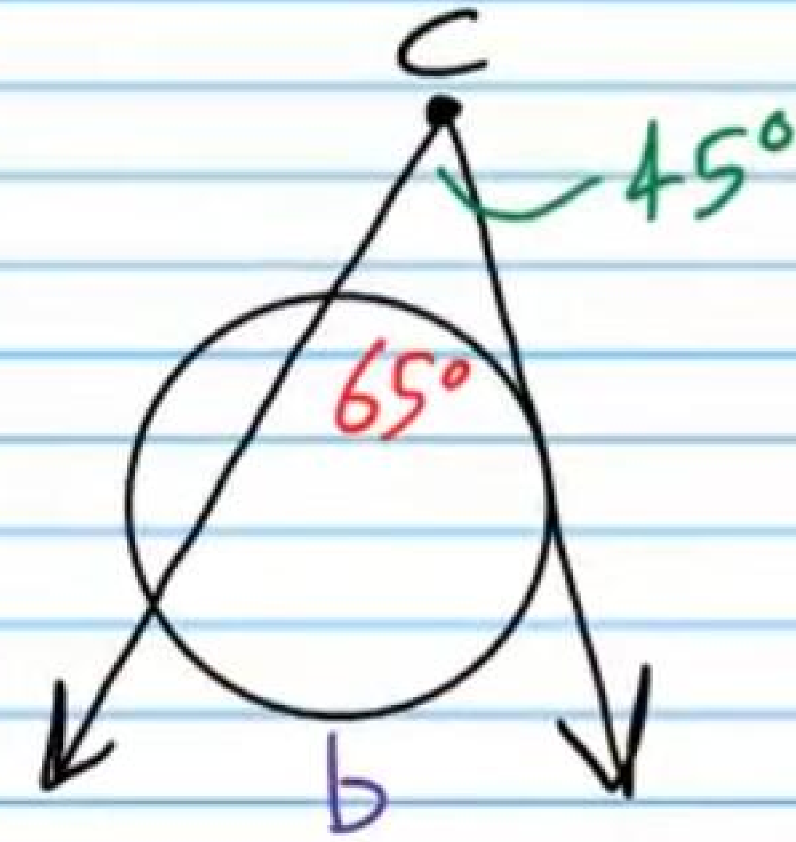
(19) Find m∠DEC.



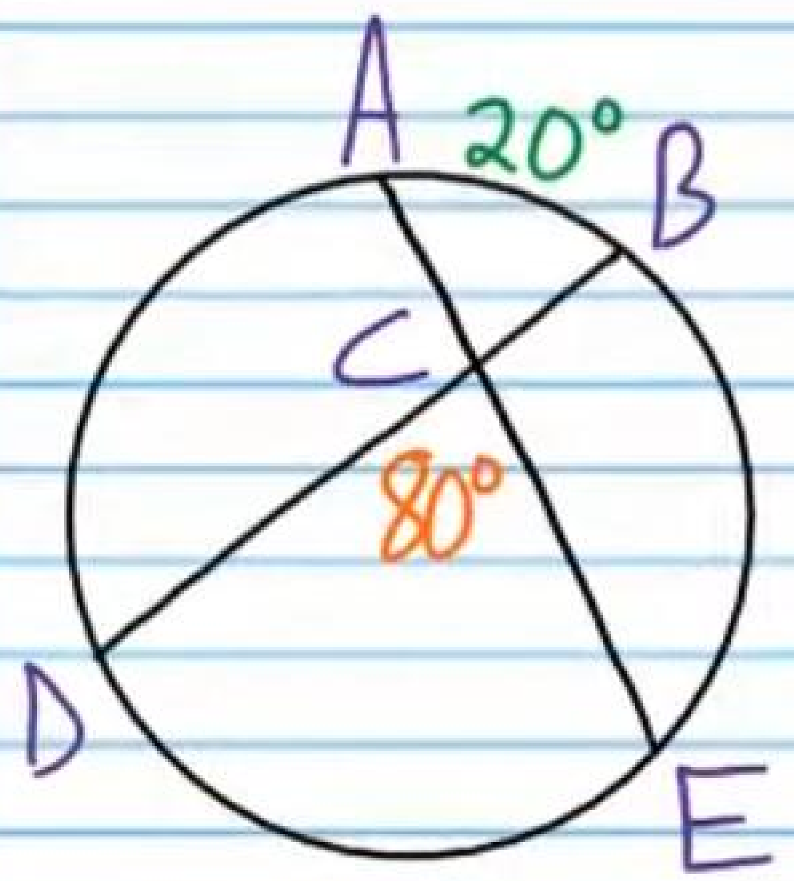
(20) Find m∠2.



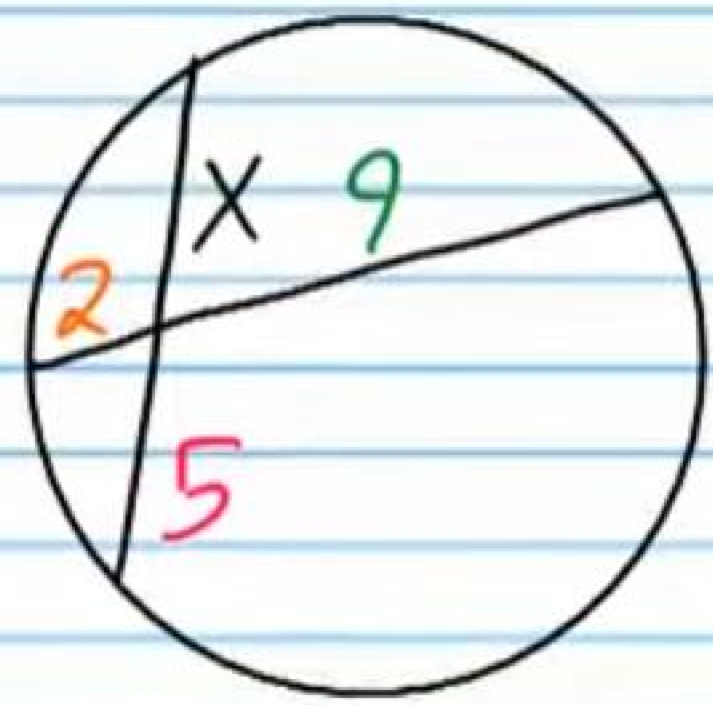
(21) Find b.



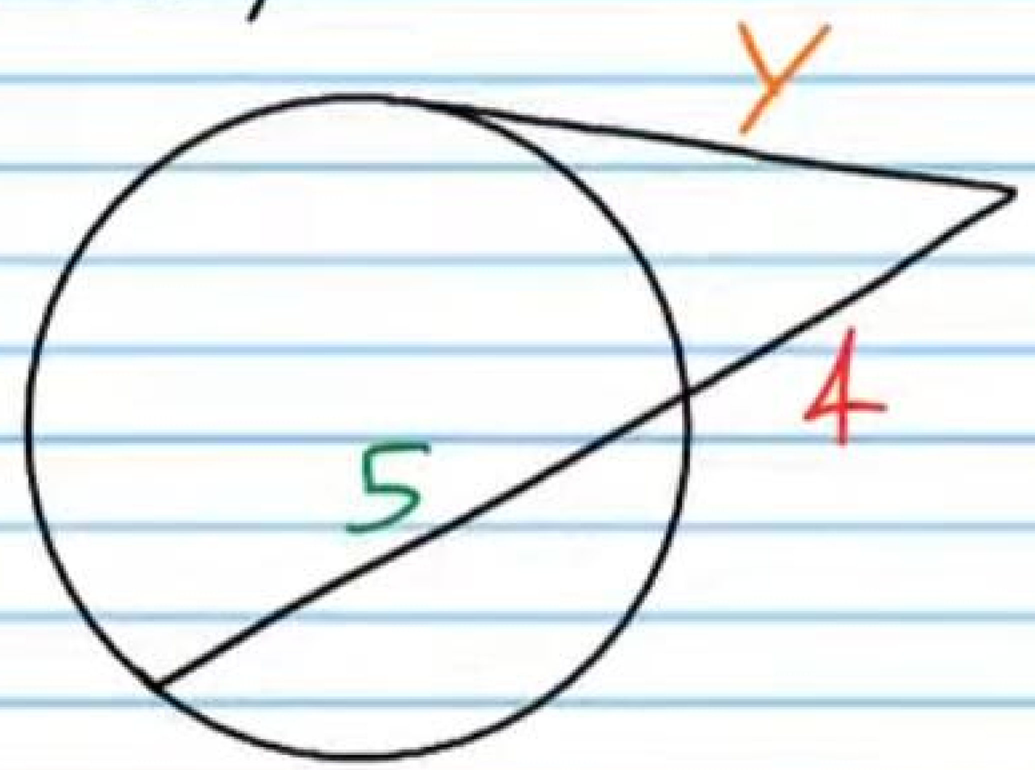
(22) Find $m\overset{\frown}{DE}$



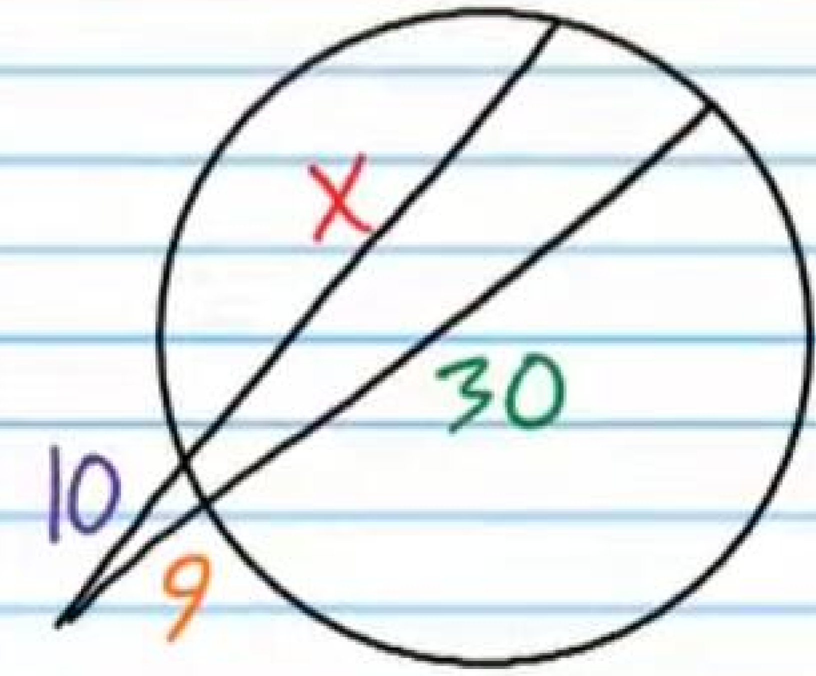
(23) Find X.



(24) Find y.



(25) Find X.



Practice Geometry Test V (Version 2) Answers

① $BC = \boxed{3}$

② i) $m\angle D = \boxed{90^\circ}$ ii) $EB = \boxed{17\text{ cm}}$

③ $m\angle AKB = \boxed{78^\circ}$

④ $\boxed{12.6\text{ m}}$

⑤ i) $m\widehat{LQM} = \boxed{180^\circ}$ ii) $m\widehat{LP} = \boxed{50^\circ}$

⑥ $x = \boxed{4\text{ cm}}$

⑦ i) $m\angle C = \boxed{90^\circ}$ ii) $m\angle A = \boxed{49^\circ}$

⑧ $\boxed{\text{No}}$ $(\sqrt{3})^2 + 1^2 \neq 2.03^2$

$3 + 1 \neq 4.1209$

$4 \neq 4.1209$

⑨ $\boxed{x = -1}$

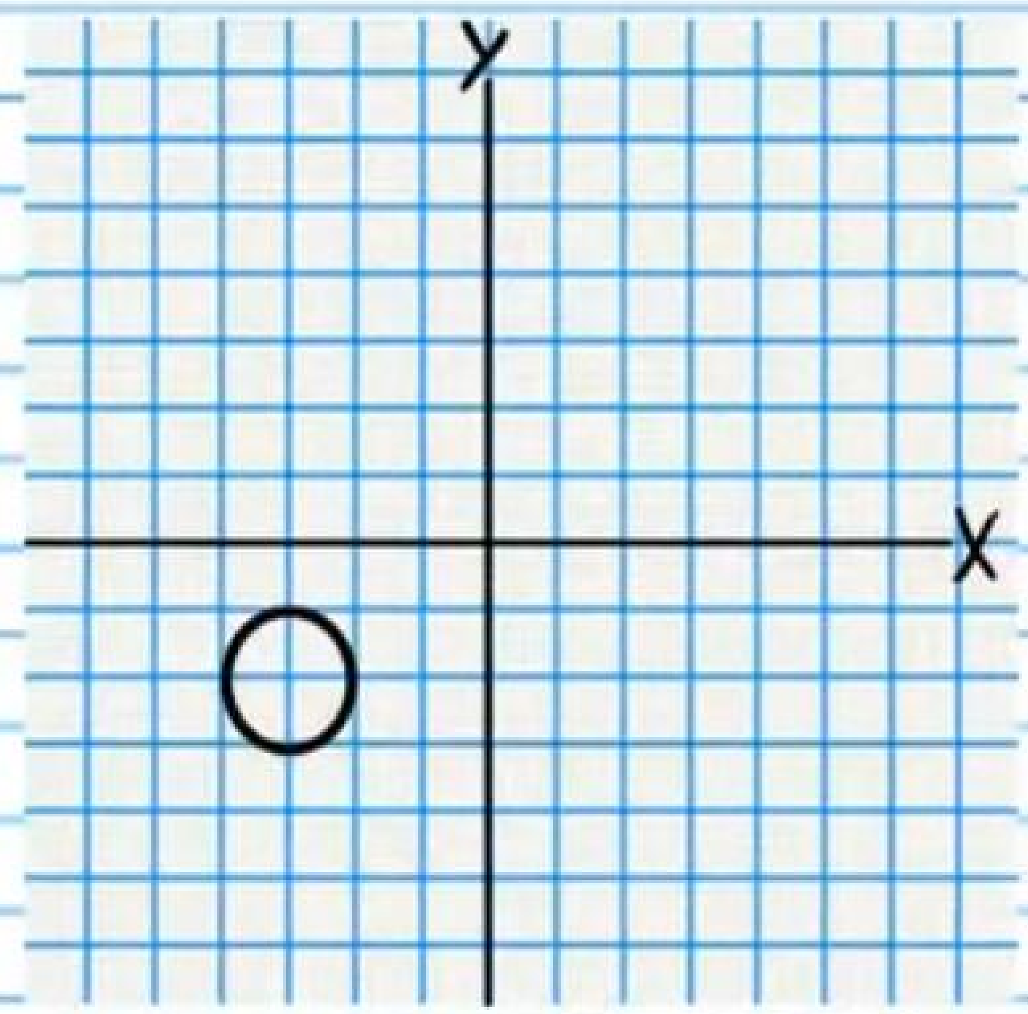
⑩ $\boxed{\text{Yes.}}$ $\overline{BC}$ is a perpendicular bisector of $\overline{AD}$, so it must be a diameter.

⑪ $\boxed{29.08\text{ m}^2}$

⑫ i) $m\widehat{HG} = \boxed{50^\circ}$ ii) $HG = \boxed{6\text{ ft}}$

⑬ $CD = \boxed{8}$

⑭ $\boxed{x^2 + (y+3)^2 = 9}$

⑮


⑯ $\boxed{(x-6)^2 + (y-1)^2 = 2}$

⑰ $m\angle E = \boxed{70^\circ}$

⑱ $m\angle C = \boxed{78^\circ}$

⑲ $m\angle DEC = \boxed{99^\circ}$

⑳ $m\angle 2 = \boxed{62.5^\circ}$

㉑ $b = \boxed{155^\circ}$

㉒ $m\widehat{DE} = \boxed{140^\circ}$

㉓ $x = \boxed{18/5}$ or $\boxed{3.6}$ or $\boxed{3\tfrac{3}{5}}$

㉔ $y = \boxed{6}$

㉕ $x = \boxed{25.1}$